
SMART TRAFFIC
MANAGEMENT
SYSTEM

Menofia Unversity
Faculty of Science
Cs Department

SYSTEM ANALYSIS & DESIGN

PROJECT

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STM
Smart Traffic Manger System



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Project Identification

- Current Problem:

Currently, traffic control in many cities is done manually or using traditional, non-smart systems. This leads to problems such as:
Severe congestion, especially during rush hour.
Slow response to traffic accidents or breakdowns.
Lack of effective coordination between traffic signals.
Lack of accurate data for prompt decision-making.
The rate of accidents increases due to haphazardness and poor control of vehicle and pedestrian traffic.

- Project Description:

This project aims to design a smart traffic management system using technology to reduce congestion, increase safety, and provide real-time control of roads.

The system will include:
Surveillance cameras to monitor traffic and accidents.
Road sensors to collect data.
A central system processes data and automatically adjusts traffic signals based on traffic pressure.
An interface for employees to monitor progress and notify the relevant authorities in real-time.
The ultimate goal: to create a system that helps improve traffic flow, reduce accidents, save people's time, and make the roads safer.

- Project Vision:

The Intelligent Traffic Management System will become a model for improving urban mobility in smart cities, serving as an integrated solution for intelligent traffic management using advanced technology. The project aims to create a safer and more efficient urban environment by improving the system's response to traffic and reducing costs resulting from congestion and accidents. This will ultimately save citizens time and reduce environmental impacts resulting from pollution.

- Project Scope:

Main Project Components:

Technologies Used:
Smart surveillance cameras to track vehicle and pedestrian traffic.
Road sensors to detect traffic and measure vehicle density.
Artificial intelligence algorithms to analyze data and determine signal timing.
A unified communication network to connect all devices and systems simultaneously.

Functional Scope:
Optimizing traffic signal timing based on live data.
Real-time incident and disruption management.
Providing accurate reports on traffic density and road conditions.
Providing an interactive interface for employees to monitor the situation around the clock.
Alert the relevant authorities when accidents or breakdowns occur.

Limitations:
The project is limited to urban areas in the first phase.
The application will be designed for major city roads.
The system's initial focus will be on improving traffic during peak hours only.
Deliverables:
Design and development of the smart system.
Installation of sensor cameras and upgrading of smart signals.
Training employees on how to use the system.
Preparing periodic reports on system performance.

- Project Objectives

Reducing congestion by improving signal and traffic flow.
Increasing safety and reducing accidents through an intelligent traffic management system.
Saving time for drivers and pedestrians by directing them to the best and fastest routes.
Data analysis to help authorities make better decisions.

System Components

1- Inputs:

The data the system needs to operate. Because the system cannot operate alone, it must have multiple data sources to analyze the situation and make decisions.
These are the things the system takes from the real world to process and produce useful results.

Main Data Sources:

• Smart Cameras (CCTV & AI Cameras):
Monitors the number of vehicles, their speed, and direction.
Detects if there is an accident or a vehicle is parked in the middle of the road.

• Road Sensors:
Counts the number of vehicles passing a certain point.
Determines the weight of the vehicles (because heavy trucks have special regulations).
• Traffic Signal Data:
Determines the current color of the signal and how long it will change.
Status of the signals (green, red, yellow).
Timing of each signal (how long it will remain on the current color).
Traffic priority (for example, main roads take longer to navigate than secondary roads).

• GPS & Maps Data:
Real-time locations of vehicles using GPS applications.
Data about moving vehicles from navigation applications (such as Google Maps, Waze).
Alternative routes for each road.
Road conditions (closed, congested, passable).

• Emergency Data:
Locations of ambulances and fire trucks.
Priority requests for ambulances, fire trucks, and police vehicles so the system can clear the way.
Fastest travel routes for emergency vehicles.

• Weather Data:
Visibility due to fog or rain.
Road conditions (dry, wet, icy).

• User Requests:
Someone looking for a parking space.
Someone wanting to know the fastest route.

2- Processing:

How does the system process data? How does the system work?
After the system collects the data (inputs), it begins processing and analyzing it to make intelligent decisions.

Basic Processing Stages:

• Traffic Analysis:
The system takes data from cameras and sensors and understands the traffic congestion on each street. It compares the current traffic congestion with previous data to determine if there is a new problem.
It determines the traffic flow speed for each road.
It detects if there is an accident or an abnormally parked car.

• Decision Making:
Does the road need to change the signal timing?
Is there an alternative route that could help reduce congestion?
Does an ambulance need to take priority?

• Smart Signal Timing Control:
It decides whether the signal timing needs to be lengthened or shortened. It prioritizes streets so that congested streets take longer to turn on the light, while empty streets take longer to turn on the light.

• Emergency Handling:
Automatically opens lights for ambulances, police, and fire trucks.
Suggests faster alternative routes away from congestion.

• Parking Management:
Monitors available parking spaces.
Sends notifications to drivers about the nearest available spaces.
It can also alert drivers if the road is congested and suggest an alternative route.
Sends alerts about accidents that may delay them.

• Traffic Prediction:
Uses artificial intelligence (AI) to predict whether the road will be congested in the next few minutes or hours.
Suggests preemptive solutions to prevent congestion.

• Real-time Data Updates:
The system compares current data to past data to learn and improve itself.

3- Outputs:

What does the system produce after the analysis?
Once the system processes the data, it begins producing results that help improve traffic.

The system produces results:

• Traffic Flow Optimization:
The system dynamically changes traffic signal timings to alleviate congestion.
It opens alternative routes and directs vehicles to them.

• Traffic Signal Management:
It changes signal timings to relieve congestion.
It prioritizes the most congested roads.

• Driver Alerts:
It sends notifications to drivers about congested and alternative routes.
It displays traffic information on road signs.

• Emergency Prioritization:
It opens signals for ambulances and fire trucks.
It suggests the best routes for emergency vehicles.

• Parking Guidance:
Provides data on available parking spaces and sends it to drivers.

• Reports & Analytics:
Provides detailed data to traffic officials in each area.
Assists the government in planning road development.

4- Performance:

How do we measure the success and efficiency of a system?
To determine if a system is successful, we must measure its performance using specific criteria.

Key Performance Indicators (KPIs):

• Response Time:
The time it takes the system to process and analyze data and make a decision.

• Traffic Flow Improvement:
Did congestion decrease after the system was activated? This means the percentage of congestion reduction compared to normal conditions.

• Prediction Accuracy:
Are the system's predictions accurate? This means the accuracy of the system's predictions about future congestion.

• Emergency Response Time:
Do ambulances arrive faster after the system is activated? This means the time it takes the system to clear the way for emergency vehicles.

• Data Update Speed:
How often is the information updated? That means every few seconds the system updates the traffic data.

5- Security:

How do we protect the system from breaches and problems? The system must be secure against cyber attacks and sabotage.

Data Security:

• Encryption:
Use encryption to protect data transmitted between sensors and servers.
All data between sensors and the system must be encrypted to prevent theft.

• Access Control:
Not just anyone can access traffic data, only authorized parties.
No unauthorized person should be allowed to access sensitive data.

• Cybersecurity Measures:
Use firewalls to prevent cyber attacks.
Install Intrusion Detection Systems (IDS) to monitor any hacking attempts.
Continuously update the system to close any security gaps.

Physical Device Protection:

• Protect cameras and sensors from theft or tampering.
• Place monitoring devices in high places to protect against vandalism.

Safety Rate & Error Rate :

Safety Rate in Traffic Management Systems
The safety rate expresses the accuracy and effectiveness of the system in preventing accidents and improving road safety. It is calculated based on the percentage of errors and problems that may occur within the system.

How to Calculate the Safety Rate (Safety Rate Formula)

- Safety Rate (%) = (1 - Error Rate) x 100

- Error Rate = (Number of accidents or errors caused by the system ÷ Total number of detected cases) x 100

A practical example for your project:

If the system monitors 10,000 cars in one day, and 50 problems occur due to system errors (such as delayed signals or incorrect decisions), the result is:
Error Rate = (50 ÷ 10,000) × 100 = 0.5%
Safety Rate = (1 - 0.005) × 100 = 99.5%
This means the system's safety rate is 99.5%, and the system operates efficiently, but there is a small error rate that needs improvement.

Factors that affect the Safety Rate:

- Sensor accuracy: If sensors are inaccurate, the error rate increases.
- Data analysis speed: If data is delayed, decisions may be incorrect.
- Cybersecurity: If the system is hacked, signals may be tampered with.
- User interaction: If drivers do not follow instructions, the error rate may increase.

If you want to calculate the Safety Rate more accurately, you can specify criteria by which to measure errors, such as signal delays, accidents, or technical failures.

Main Analysis

1- Business Function:

The basic functions or tasks that a system performs to achieve its objectives. These are simply the tasks the system performs to solve the primary problem for which it was created.
What is the business function of a traffic system?
In other words, why does the traffic control system work? What does it do to improve traffic? What benefits does the system offer to users and traffic authorities?

Main functions of the system:

• Smart Traffic Light Control:

Adjusting signal timings based on congestion.
Giving priority to emergency vehicles (ambulances, fire engines, and police).
Improving traffic flow and reducing waiting times at signals.

• Road Monitoring & Data Analysis:

Using cameras and sensors to monitor vehicle movement and identify congested streets.
Identifying accidents and congestion and proposing immediate solutions.
Analyzing traffic patterns to assist relevant authorities in future planning.
It counts the number of cars on each street and identifies accident locations.
It sends alerts if there is an abnormality (for example, a broken-down car or an accident).

• Emergency Handling:

Facilitates the movement of ambulances and fire trucks by automatically adjusting signals.
Sends alerts to drivers about nearby accidents.
Gives priority to emergency services on busy streets.

• Smart Parking Management:

Identifies available parking spaces and sends their data to drivers.
Manages smart parking spaces and analyzes their usage.
Reduces congestion resulting from searching for parking spaces.

• Real-time Traffic Information:

Sends drivers live updates on road conditions.
Suggests alternative routes to avoid traffic jams.
Integrates with GPS applications such as Google Maps and Waze.

• Decision Support for Authorities:

Provides data and analysis on the most congested areas.
Assists governments in planning new road networks.
Providing reports on the causes of accidents and traffic congestion.

Summary:

The business functions of a traffic management system include:

- ✅ Intelligently regulating traffic signals
- ✅ Traffic monitoring and analysis
- ✅ Improving emergency response
- ✅ Parking management
- ✅ Providing real-time information to drivers
- ✅ Supporting governments in planning and decision-making

2- Strategic Objectives :

The strategic objective is the strategic goal or overarching purpose that the system seeks to achieve in the long term. It's not just a simple improvement; it's the overarching goals the system seeks to achieve in the long term.

The strategic objectives of an intelligent traffic management system:

• Reducing traffic congestion :

Using artificial intelligence to intelligently manage signals and reduce bottlenecks.
Providing alternative routes for drivers in real time.
Improving vehicle flow in vital areas

• Improving emergency response :

Allowing ambulances and fire trucks to pass quickly through intelligent signal control.
Reduce emergency response times and save more lives.

• Reducing accidents and improving traffic safety :

Monitoring traffic violations such as speeding or running red lights.
Alert drivers to road conditions (e.g., "slippery road due to rain").
Analyzing accident data to help relevant authorities improve roads.

• Reducing environmental pollution :

Reducing waiting time at signals reduces fuel consumption and carbon emissions.
Reducing the need to stand for long periods in traffic jams.

• Improving driver experience :

Providing real-time information on traffic congestion and available parking spaces.
Reducing daily commute times and making driving smoother.

• Supporting future smart city planning :

Collecting traffic data to help governments plan more efficient infrastructure.

Supporting future projects such as self-driving cars and smart roads.

3- Initial Cost

The initial cost is the initial cost of establishing and operating the system for the first time. This includes all the expenses you will incur to start the project before it actually goes live. It's the money we will spend initially to prepare the system and start it working, before we start seeing its benefits.

Initial Costs for a Smart Traffic Management Project:

- **Hardware Cost:**

Surveillance cameras to monitor roads and detect congestion and violations.
Sensors to measure the number of cars and traffic conditions.
Powerful servers to store and process traffic data.
Traffic signal controllers to connect them to the smart system.

- **Software Development Cost:**

Design and programming the software that will operate the system and monitor signals and traffic congestion.
Artificial intelligence and data analysis to calculate congestion and adjust signals, helping with quick decision-making.
Developing an application for drivers to provide them with real-time road data.

- **Installation & Deployment Cost:**

Installing cameras and sensors on the streets.
Connecting devices to each other and connecting them to servers.
Testing the system and ensuring it is operating efficiently.

- **Maintenance & Support Cost:**

Updating and maintaining the system regularly.
Technical support to resolve any issues faced by users or relevant authorities.

- **Training & Awareness Cost:**

Training those responsible for operating the system.
Educating drivers on how to use the system to benefit from it.

4- Transaction Process

Simply put, it's the steps any event or process goes through within the system from start to finish. That is, when something happens in traffic, how does the system handle it?
It's the way the system handles any event or situation that occurs in traffic, step by step.

How does the system handle the processes?

Example: A car approaches a smart traffic light

- The system collects data:

Sensors and cameras detect cars stopped at the light.
The number and speed of cars are recorded.

- Data analysis:

The system compares the traffic on this street with other streets.
It determines whether or not to extend the green light.

- Decision making:

If there's traffic, the system extends the green light.
If the road is empty, the light stays on for the shortest possible time.

- Decision execution:

The light changes based on the analysis.
A notification is sent to drivers in the app about the traffic situation.

- Transaction Recording:

The system saves data so it can analyze overall traffic flow.
A report is sent out detailing the number of vehicles and traffic congestion at the time.

Simply put, the system does four things:

- It collects data from cameras and sensors and identifies the problem (for example, traffic congestion at a traffic light).
- It analyzes the situation, determines the appropriate solution, and makes a decision (make the green light longer or shorter).
- It executes the decision (changes the signal).
- It stores the data for later use in improving traffic.

This means that the system operates 24/7, making intelligent decisions in real-time, while simultaneously collecting data useful in the future. This means that the system operates automatically, makes decisions quickly, and learns from the data it collects.

Types of System Operations:

- Real-time Transactions:

Immediately changing signals based on traffic congestion.
Opening the road for ambulances and fire trucks.
Alerts drivers to traffic conditions in real-time.

- Processes that require storage and analysis (Stored Transactions):

Storing traffic data for later analysis.
Calculating daily and weekly traffic averages.
Submitting reports to relevant authorities to improve traffic.

4- Information Needed

What information does the user need?
Users need information to help them make decisions while driving or managing traffic.

1- Information for Drivers

Where is the traffic jam? (Is the road open or is there traffic?)
The best alternative route if the road is congested?
When will the traffic light open? (How many seconds until it turns green?)
Empty parking spaces so you don't waste time searching.
Important warnings (accident, road closure, maintenance work).
Is there a violation? (If you ran a red light or speeded too much).

2- Information for the Traffic Department

The number of cars on each street so they can identify congested areas.
Congestion analysis (when the roads are most congested).
Vehicle speed (if someone is driving at a dangerous speed).
Accident locations so they can dispatch an ambulance or police quickly.
Traffic violations (who ran a red light or speeded).

Simply put, how does this information help?

The driver chooses the right route and doesn't delay.
The traffic department knows how to control traffic signals to reduce congestion.
The system reduces accidents and keeps roads organized.
The system collects this information to make driving easier and traffic flow better!

6- Outputs

Outputs are the results the system produces after processing the data. These are the things users see or benefit from.

1. Outputs - For Drivers

Real-time traffic conditions (e.g., "Corniche Road is congested, use an alternate route").
Remaining signal time (e.g., "The signal will turn green in 10 seconds").
Best alternate route if there is traffic.
Available parking spaces.
Emergency alerts (such as an accident or a closed road).

Violation messages (e.g., "You have been issued a speeding ticket").

2. Outputs - For Traffic Management

Congestion reports (number of cars in each area and the most congested streets).
Peak time analysis (the times when the streets are most congested).
Violation records (who ran the signal, who speeded).
Accident statistics (number of accidents and their locations).
Recommendations for improving traffic (for example, "It's best to adjust the timing of the signals on a particular street").

Put simply, what do the outputs help with?

It helps the driver choose the right route and avoid delays.
Traffic can intelligently control signals to reduce congestion.
The system reduces accidents and better organizes roads.

This means that the system not only collects data, but also generates useful information for all users!

Analytical questions

1- Who ?

Who will use the system?
Drivers: To know traffic conditions, less congested roads, and parking spaces.
Traffic Management: To monitor streets, adjust signals, and reduce accidents.
Police: To monitor accidents and violations.
Ambulance and fire department: To reach emergency services as quickly as possible.
Pedestrians: Because better traffic management will make crossings safer.

Why ?

Why is the system useful for all of these groups?
Because it helps each of them make faster decisions:

📌 Drivers will know how to choose the right route and save time.

📌 Traffic management will be able to control traffic efficiently.

📌 Police will be able to monitor violations easily.

📌 Ambulance and fire department will reach emergency services without delay.

📌 Pedestrians will be able to cross the streets more safely.

2- What ?

What is the system?
The system is a smart system that regulates traffic using cameras, sensors, and artificial intelligence to reduce congestion, make roads faster and safer, and improve vehicle flow.

Why?

Why do we need a system like this?

📌 Because regular roads are congested and delayed.

📌 The traditional system is inflexible and unresponsive to actual traffic conditions.

📌 The smart system will help reduce accidents and improve traffic management.

3- How?

How will the system work?

It collects data from cameras and sensors about the number of vehicles and traffic congestion.
It analyzes traffic and decides to adjust signals or suggest alternative routes.
It sends signals and alerts to drivers via smart panels or an app.
It records data to analyze traffic more broadly over time and improve itself over time.

Why?

Why does the system need all these steps? Why does it need to work this way?

📌 To be able to make accurate decisions based on actual data, instead of relying on fixed signal timings without flexibility.
📌 To be flexible and adapt to any changes in traffic.
📌 To maintain a record of traffic and violations for the future.

4- When?

When will the system operate?

The system will operate 24 hours a day, non-stop, but it will be most active during peak times when traffic is at its highest.

Why?

Why does it need to be running all the time?

📌 Because traffic doesn't just happen at certain times.
📌 Accidents and emergencies can happen at any time.
📌 It must always be ready to organize and improve traffic.

5- Where?

Where will the system be implemented?

It will be implemented in crowded cities, especially in areas with heavy traffic, major intersections, or areas near schools and hospitals.

Why?

Why not everywhere?

Because priority is given to areas with a real problem, and after its success, it may be expanded to include more areas.

Fact Finding

1. Interviews

Interview people involved in the system and ask them about traffic problems and how to improve them.

Who are you targeting?

Drivers: Ask about the areas where they encounter the most traffic congestion and problems.
Police and traffic police: Identify the areas with the most accidents and traffic delays.
Citizens and pedestrians: Check whether the signals help them or cause more congestion.

Interview 1: (with Traffic police Officer)

1- How're you?

- I'm fine, thank you.

2- What are the main traffic issues?

- The main traffic issues in Egypt include severe congestion, especially during peak hours, leading to long delays. The roads also suffer from frequent accidents due to speeding and unsafe driving behaviors. Traffic signal management can be ineffective, increasing wait times at intersections.

3- Are there certain times and locations of day when traffic control needs to be more intensive?

- Yes, traffic control needs to be more intensive during morning and evening rush hours, particularly in high-traffic areas like business districts, major intersections, and near schools. Special events or roadwork situations also require increased monitoring.

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4- How would you like to see the success of the system measured?

- Reducing congestion and waiting time in traffic, thus reducing travel time and reducing accidents.

5- What features would you find most beneficial in a smart traffic management application?

- Improving traffic movement and predicting congestion by providing information to drivers and detecting violations through the use of smart cameras, improving safety and improving the response of emergency services by identifying faster routes, improving drivers' behavior, reducing carbon emissions by improving traffic movement and reducing congestion, and obtaining more accurate information through Connecting the system to sensors.

6- **Who has permission to use the application?**

the government
traffic police
Some transportation companies
Drivers

7- How do you see the future of AI in traffic management?

- I believe AI will bring a transformative shift in traffic management in the near future. With advancements in technology, smart traffic systems will be able to improve vehicle flow, reduce congestion, and even predict accidents before they happen. We hope AI technologies will help provide safer, more efficient roads for citizens and enhance the overall driving and commuting experience.

8- What type of data do you want citizens to enter when registering?

- We require some basic data from citizens, including full name, national ID number, phone number, and email address. Additionally, they are asked to provide information about their vehicle type and license plate number. This data helps us accurately identify them.

9- What are the factors that hinder the development of the violations system?

There are several factors that hinder the development of the violations system, with the primary ones being the lack of accurate, comprehensive data and limited awareness of modern technologies. There are also challenges in training staff to use these technologies. .

10- Do you think the application will succeed?

Yes, I think so.

11- What do you think of success rates?

80%

12- **What technologies are currently used to monitor traffic conditions؟**
Smart Cameras and video surveillane
Road sensors
GPS

13- **How is traffic congestion handled in emergency or peak times?**

Manual control of traffic lights: to effectively direct cars .
Digital signs have been used on roads to advise drivers about accidents

14- **Do you think that the traffic light system would work better if it was controlled by artificial intelligence to solve the congestion problem, or is the human element better?**

Yes, it has many advantages compared to the human system, and it may be more efficient in improving traffic flow, high response speed, and speed of dealing with emergency circumstances.

15- **What will the system do for the users?**
-reduce traffic congestion and make roads safer
-help drivers follow traffic rules
-collecting and analyzing accurate data
-help plan better infrastructure

16- **How do you want the external appearance of the application?**
An easy interface for the user to deal with the system

17**- Do you want to add anything to the application?**
No,Thank you.

Interview 1 Report

Person Interviewed:	**Traffic police officer**
Interviewer:	Riyad Ayman Aya Eisa Zeyad Mohamed Abd Allah Mohamed
Purpose of Interview:	1. Determine System's Details.
Summary of Interview:	1-**Traffic Issues**: Severe congestion, accidents caused by speeding, and poor traffic signal management, especially during peak hours. 2-**Smart Traffic Management Features**: Reducing congestion, improving safety, and detecting violations using smart cameras. 3-**Users and Data**: The application is aimed at the government, traffic police, transport companies, and drivers; it requires basic registration data. 4-**AI's Role in Traffic Management:** AI will improve traffic flow, reduce congestion, and predict accidents. 5-**Handling Congestion and Emergencies:** Manual control of traffic signals and the use of digital signs to guide drivers.

Interview 2
(With Director of the General Traffic Department)

1. **What are the main challenges you are currently facing in managing traffic violations 1**
   "We are facing several challenges the most prominent of which are the difficulty in accurately tracking violations, delays in updating data in a timely manner, and dealing with the large volume of violations manually, which creates an additional burden and increases the likelihood of errors."

2. **How are traffic violations currently recorded and monitored? Are there difficulties in this process?**
   "Currently, violations are recorded manually on paper or using outdated systems, which leads to slow processes and increased chances of errors. Additionally, retrieving data becomes complicated, which affects the speed of decision-making."

3. **What errors or problems would you like to avoid when developing the new system?**
   "The main issues we want to avoid are repeated errors such as entering data multiple times, data loss, and delays in updating records. We need a system that ensures data accuracy and real-time updates."

4. **How can the new system contribute to improving the experience of handling traffic violations?**
   "We aim for the new system to allow real-time recording of violations, along with automatic notifications to violators about outstanding violations. We also want integration with electronic payment systems and the ability to generate detailed reports that help analyze patterns and trends."

5. **Would you like to track additional data such as the locations of violations and the times they occur?**

"yes, We would like the system to track more detailed data, such as the locations of violations and the times they occur, which will help us analyze patterns and the circumstances surrounding the violations in order to identify areas that need enhanced monitoring and improve overall performance

6. **Does the new system require different levels of permissions for users?**
   "Yes, the new system will require different permissions for each type of user. For instance, officers will need permission to register violations only, while administrators will need access to manage the database, and drivers will need permission to view and pay for violations."

7. **What criteria should be considered when designing the user interface?**

   "The system should be easy to use and feature an intuitive interface for users. It is essential that notifications are clear and direct without complications, and that the interface supports smooth interaction with the data, allowing drivers to quickly and easily access the necessary information. Additionally, the system should be compatible with all mobile devices to provide a convenient user experience."."

8. **What are the most common complaints you receive from users (officers, administrators, or drivers) about the current system's interface?**

The most common complaints about the current system's interface are its slow performance, lack of intuitiveness, and difficulty in managing data. Officers find it inefficient for recording violations, administrators struggle with data management due to poor navigation and filtering, and drivers are frustrated by slow load times and unclear payment instructions. Overall, users find the system complex and not user-friendly, which hampers workflow and increases errors.

9. **You mentioned that there are consequences resulting from the delay in updating violation data. What are these consequences? Can you provide an example of how this affects decision-making or execution?**

Delayed updates on violation data lead to inaccurate decisions, such as imposing fines based on outdated information, causing confusion among drivers. It also hinders officials' ability to monitor violation patterns and take timely actions to improve enforcement in high-violation areas

10. **How do you expect this system to affect revenue and costs?**

    "We believe that the new system will help reduce costs in the long term by improving the management of violations, increasing efficiency in the payment process, and reducing accidents. Moreover, innovative services such as real-time notifications will attract more customers, contributing to increased revenue. Providing valuable services like route suggestions could also enhance customer loyalty and reduce turnover rates."

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Interview 2 Report

Person interviewed	Director of the General Traffic Department
Interviewer	Ziad Mohamed Riyad Shoura
Purpose of interview	To gather insights on current challenges and desired features for a new traffic violation management system.
Summary of interview	Current Challenges: ●Difficulty in tracking and updating violations ●Manual processes are slow and error-prone Desired Features: ●Real-time violation recording Automated notifications and payments Detailed reports ●User-friendly interface for all users Benefits: ●Improved efficiency and accuracy Reduced costs and increased revenue ●Enhanced customer satisfaction ●Data-driven decision-making

Interview 3 (With Users)

1.What are the most common reasons that you think cause traffic congestion in your area?
The common causes include poorly timed traffic signals and a high density of vehicles during peak hours

2. What are the biggest obstacles you face when trying to avoid traffic congestion? How can the new system help?
The biggest obstacles are not knowing the best alternative routes or the precise timing of congestion. The system can solve this problem by providing accurate data and continuous updates..

3. How would you describe your experience with current traffic signals?
Sometimes they work efficiently, and other times they are irregular, causing chaos.

4 You mentioned poor traffic signal timing as a problem. Can you explain specific scenarios where this caused issues?

During rush hours, at busy roads where signals don't adapt to real-time traffic conditions, leading to long wait times on some lanes and congestion spillovers.

5 What type of information do you need while driving to reduce risks and improve your experience?
I need information about road conditions, speed limits, camera locations, and any changes in traffic flow.

6. How do you see technology's role in reducing frequent accidents, and what features do you expect in the system to achieve this?
Technology can reduce accidents by monitoring unsafe behaviors like speeding and providing drivers with instant reports about potential road dangers.

7. Do you think automatic violation recording will positively impact drivers' behavior? Why?
Yes, because automatic recording reduces the chance of manipulation or neglect and makes drivers more careful to follow the rules.

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8 . **Would you be willing to use a system that provides real-time traffic updates, even if it involves a subscription cost?**

Yes, because the system could help me avoid traffic jams and find faster routes,

which would save time and reduce fuel costs. In the long run, this could save me more money than the subscription fee.

9. What are the main challenges that the new system might face?
The main challenges include lack of understanding of the system, cost, and resistance from drivers.

10. What advice would you give to ensure drivers accept the new system?
Use a simple interface, voice commands, and clear, quick notifications that do not distract drivers.

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Interview 3 Report

Person interviewed	**Road users**
Interviewer	**Abd Allah Mohamed** **Aya Eisa**
Purpose of interview	**Gather users' insights and needs about traffic problems.**
Summary of interview	**1. Traffic congestion is caused by poorly timed signals, high vehicle density, and irregular traffic lights.** **2. Common wrong behaviors include speeding, running red lights, and unsafe driving.** **3. Users have difficulty finding alternative routes and predicting congestion times.** **4. Drivers need information on road conditions, speed limits, camera locations, and traffic updates.** **5. The new system may face challenges like high costs and lack of understanding but it should have a simple interface and clear notifications for acceptance.**

2. Questionnaires

The idea: Design an electronic or paper survey and ask a large number of .people about their opinions on traffic and the problems they face

Questionnaire 1 (with Users)

Name *

Your answer

gender *

- male
- female

age *

- 18-30
- 30-40
- 40-50
- 50-60

Mode of transportation *

- car
- motorcycle
- public transport
- bicycle
- walking

Have you ever been late for a long appointment due to poor traffic management *

- ◯ Yes
- ◯ No

Do you prefer to use a mobile application or a website ? *

- ◯ mobile application
- ◯ website

Have you ever used online applications related to the traffic system before

- ◯ Yes
- ◯ No

How do you prefer to pay fines? *

- ◯ online
- ◯ offline

How often do you encounter traffic congestion during your daily commute? *

- ◯ always
- ◯ sometimes
- ◯ usually
- ◯ never

How does traffic congestion impact your daily life? *

- ☐ time wasted
- ☐ increased stress
- ☐ fuel consumption

What are the primary causes of traffic congestion in your area? *

- [ ] road construction
- [ ] accidents
- [ ] rush hour
- [ ] poor traffic management

How do you expect the intelligent traffic management system to affect your daily routine? *

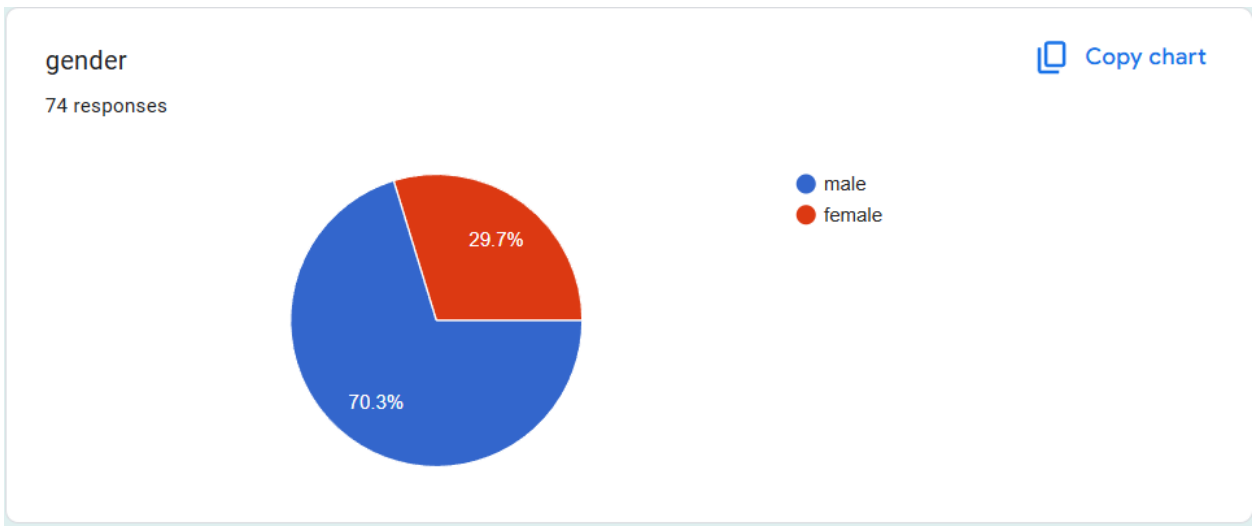
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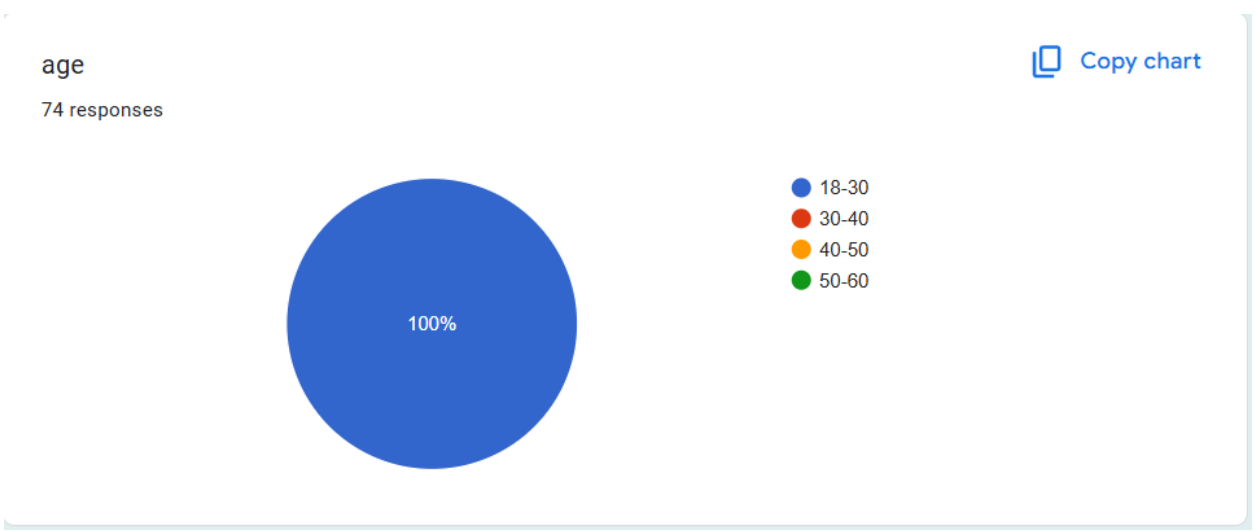
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What are your suggestions for improving traffic flow and reducing congestion in your area?

Your answer

Results and StatisTics

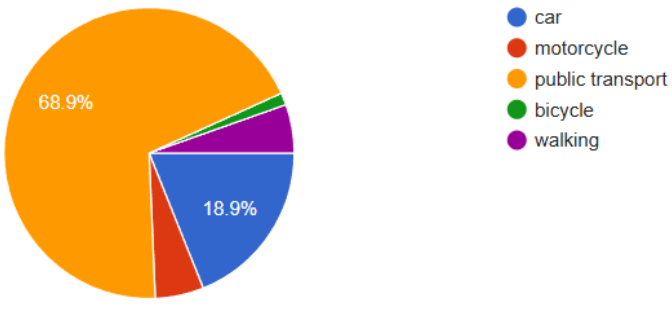




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Mode of transportation

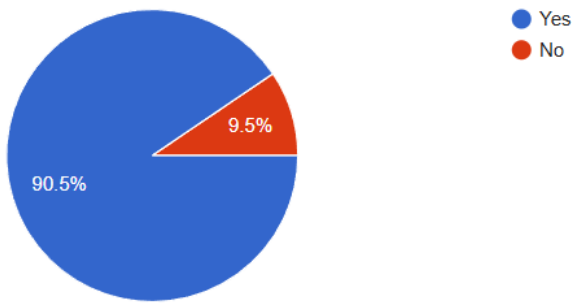
74 responses



Copy chart

Have you ever been late for a long appointment due to poor traffic management

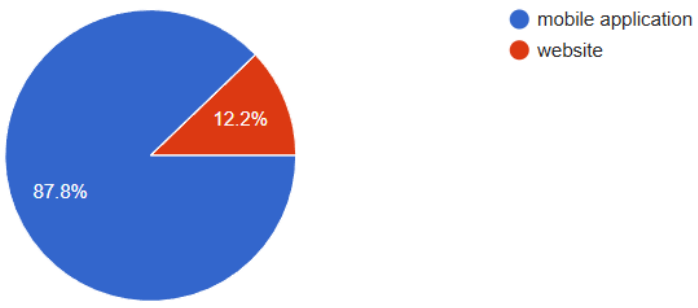
74 responses



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Do you prefer to use a mobile application or a website ?

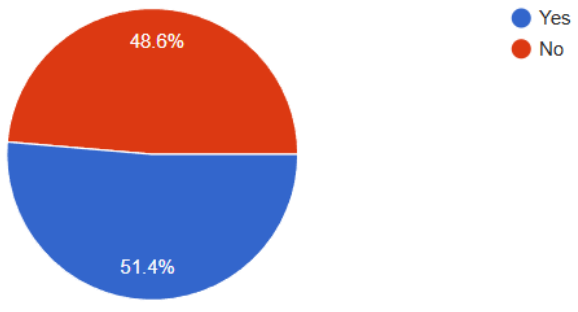
74 responses



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Do you have concerns about privacy and data collection related to your movement while driving?

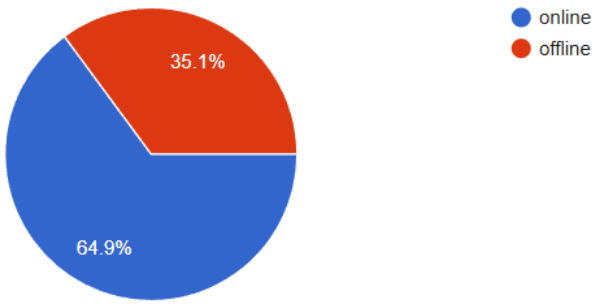
74 responses



How do you prefer to pay fines?

Copy chart

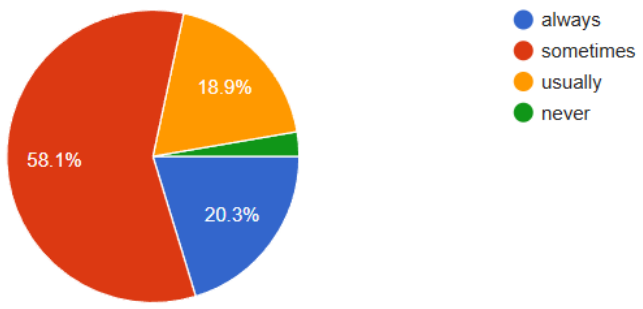
74 responses



How often do you encounter traffic congestion during your daily commute?

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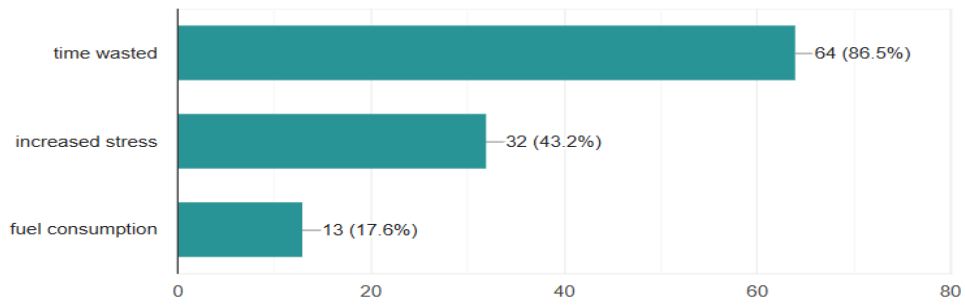
74 responses



How does traffic congestion impact your daily life?

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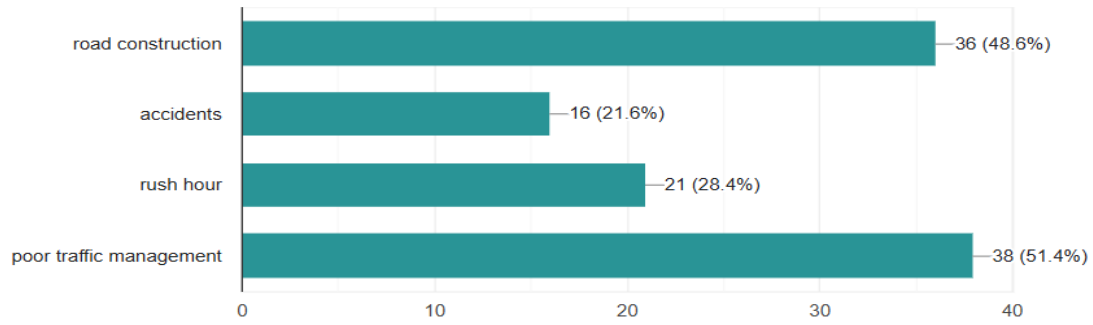
74 responses



What are the primary causes of traffic congestion in your area?

Copy chart

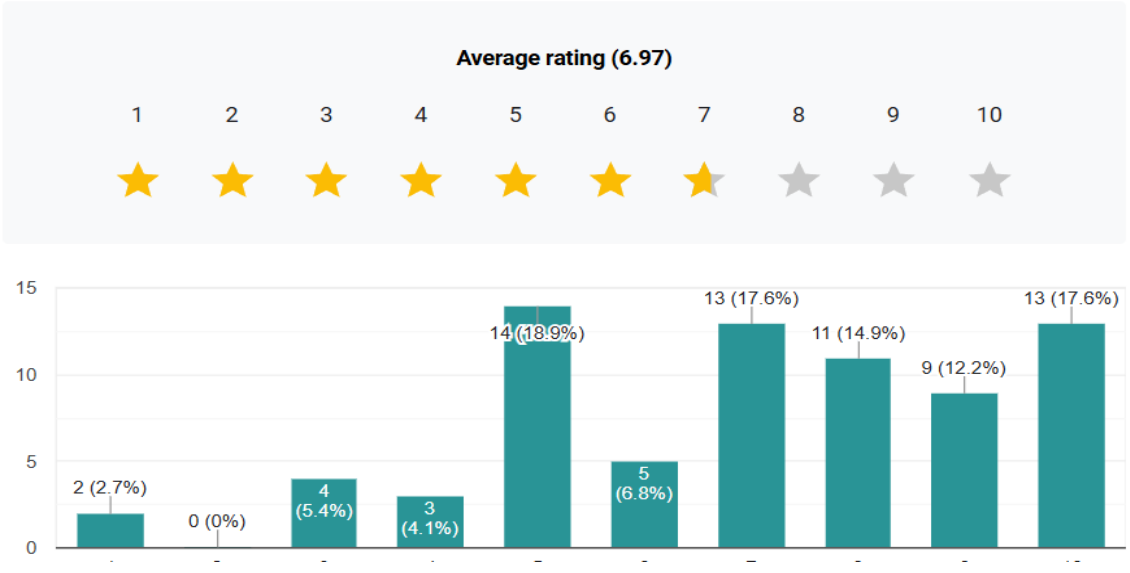
74 responses



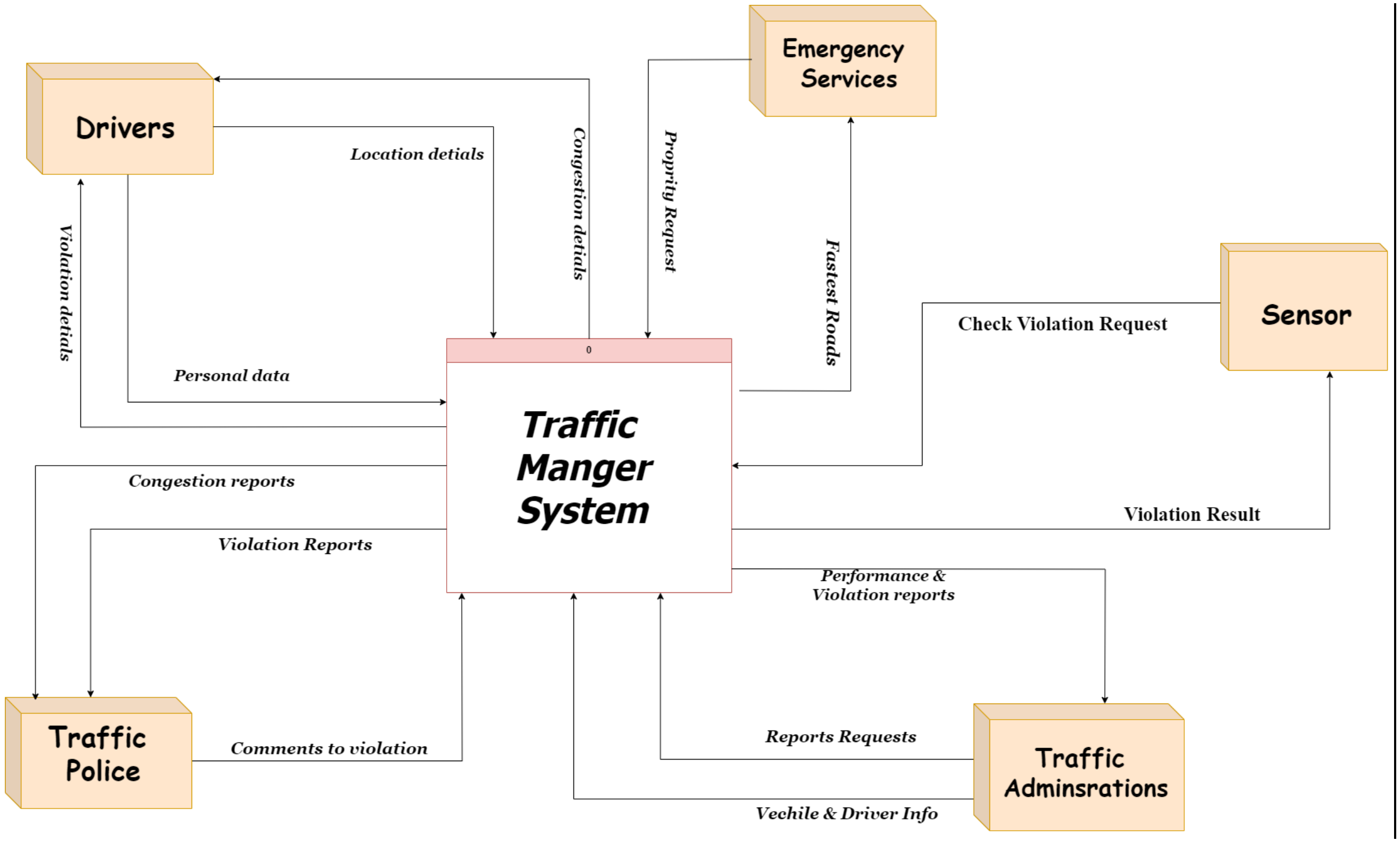
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How do you expect the intelligent traffic management system to affect your daily routine?

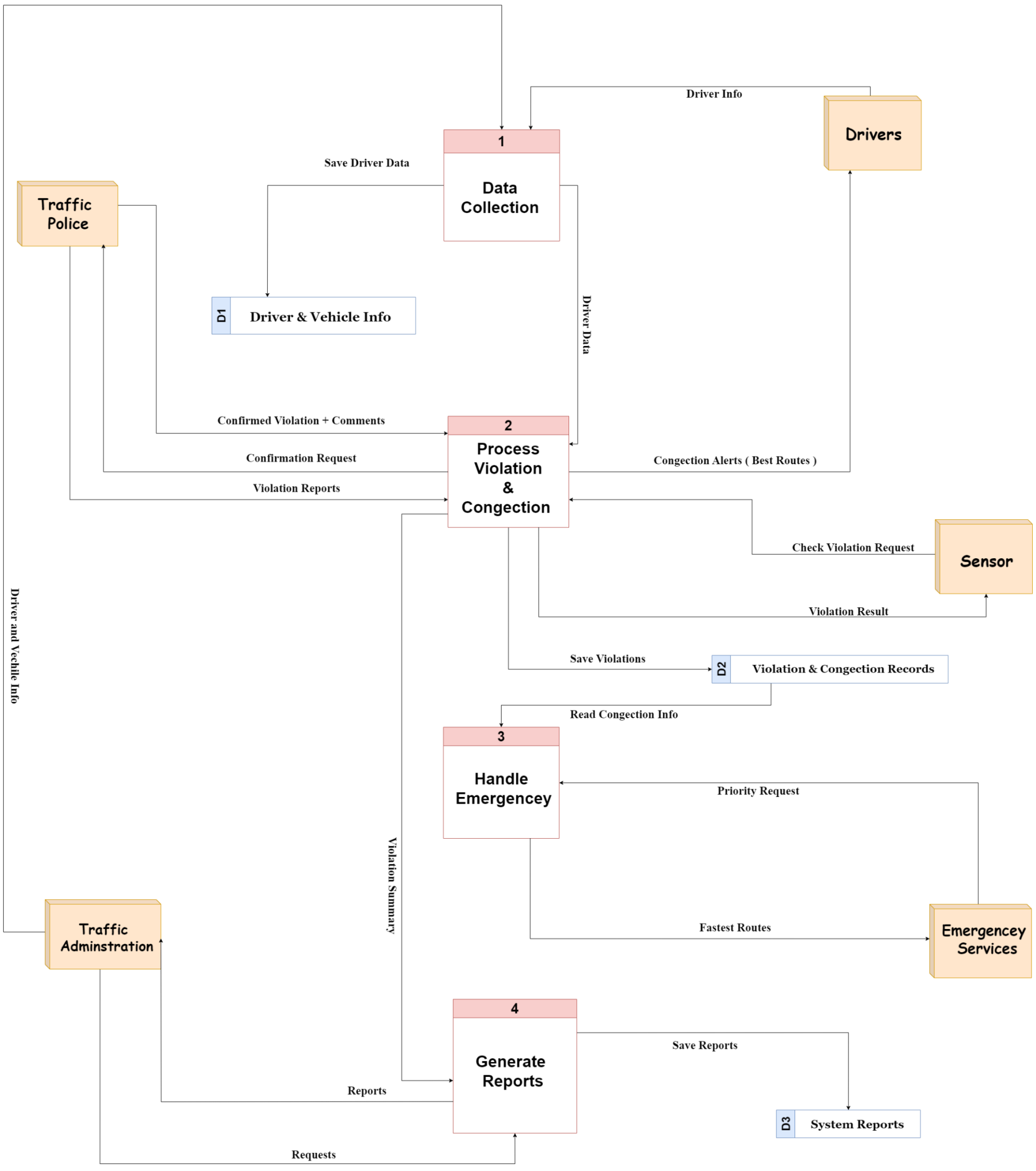
74 responses



Context Diagram

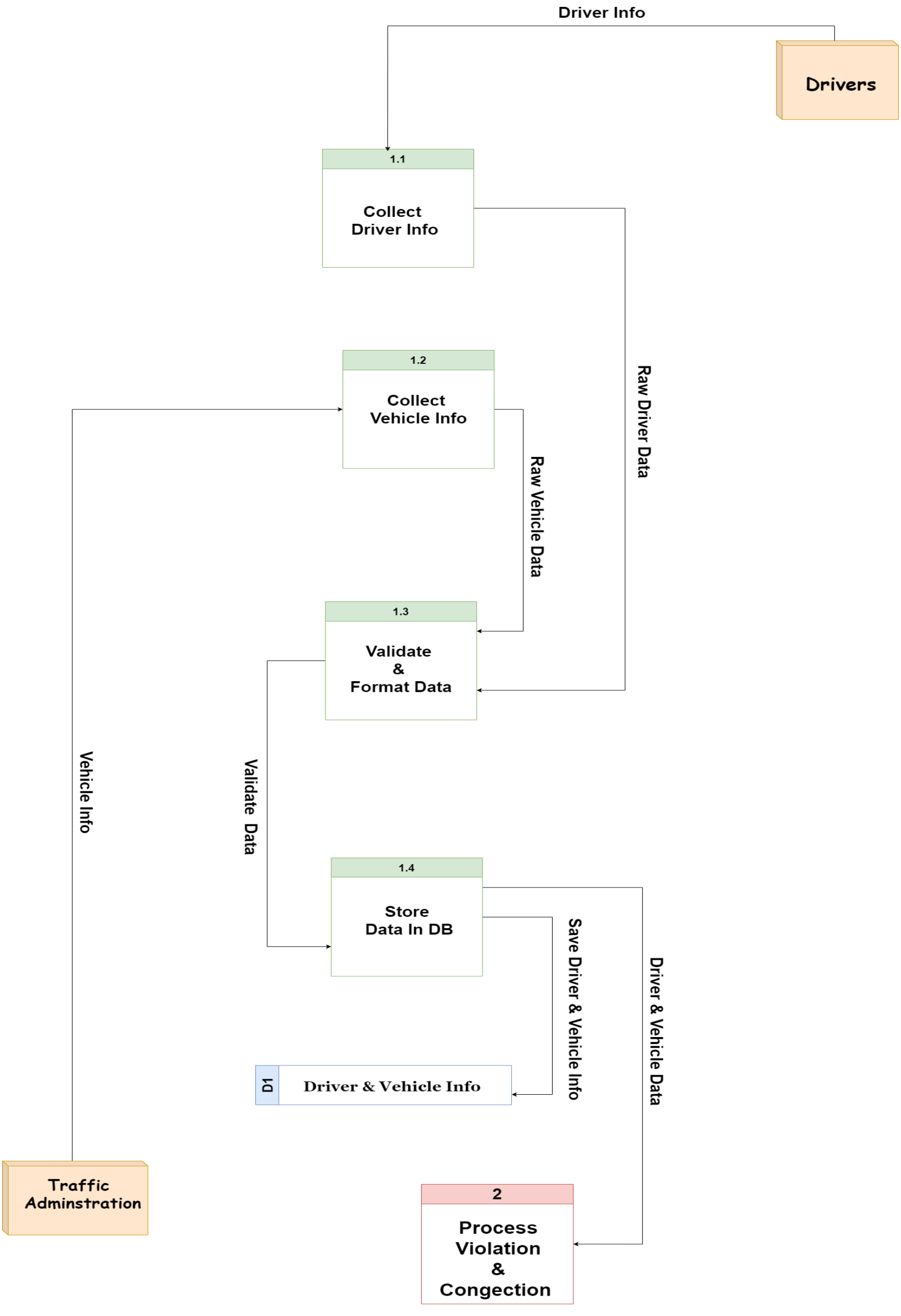


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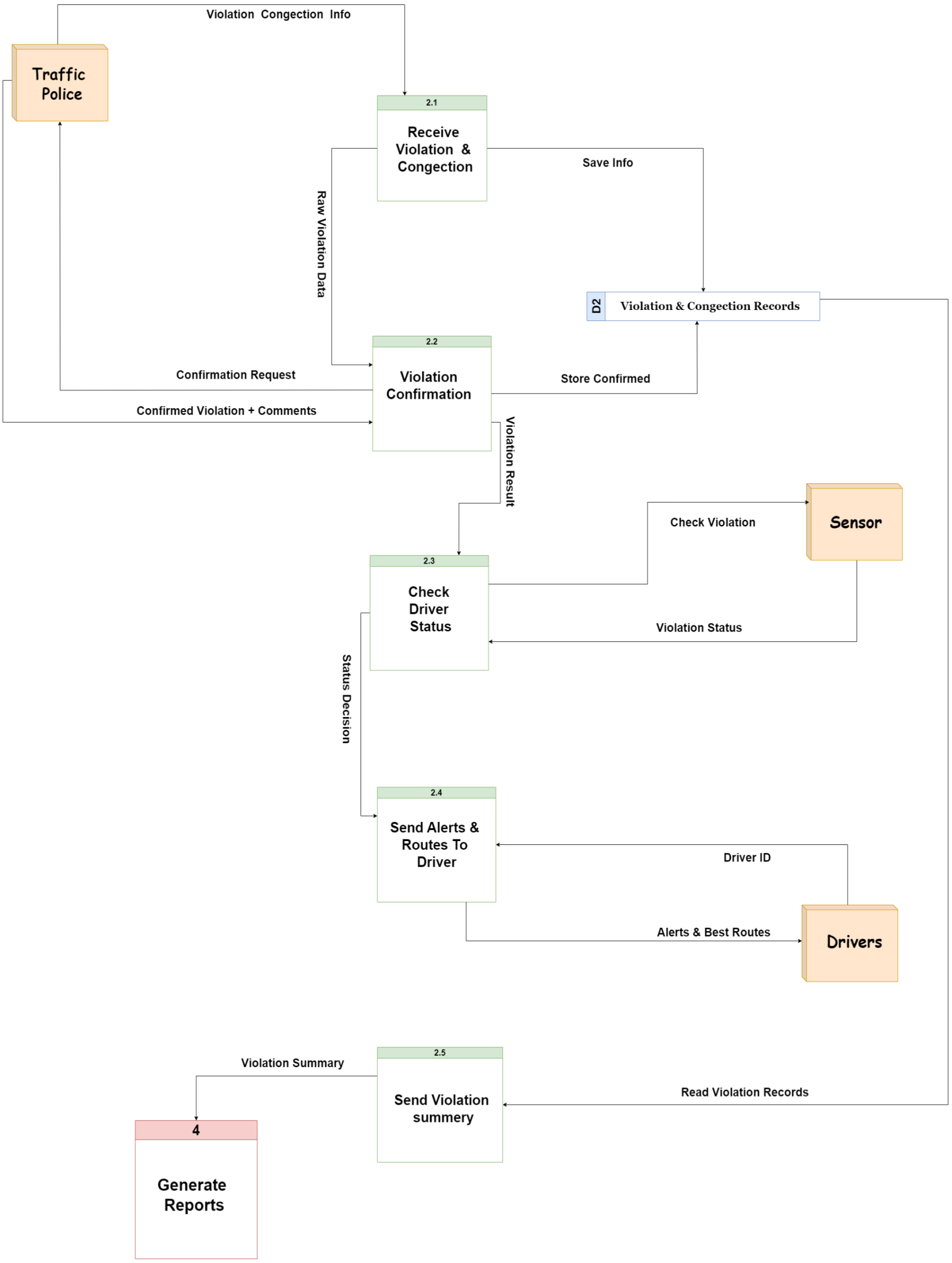

Driver Info
Drivers
1
Save Driver Data
Data Collection
Traffic Police
D1
Driver & Vehicle Info
Driver Data
Confirmed Violation + Comments
2
Process Violation & Congection
Confirmation Request
Congection Alerts (Best Routes)
Violation Reports
Check Violation Request
Sensor
Violation Result
Driver and Vechile Info
Save Violations
D2
Violation & Congection Records
Read Congection Info
3
Handle Emergencey
Priority Request
Violation Summary
Traffic Adminstration
Fastest Routes
Emergencey Services
4
Generate Reports
Save Reports
Reports
D3
System Reports
Requests

Lower level

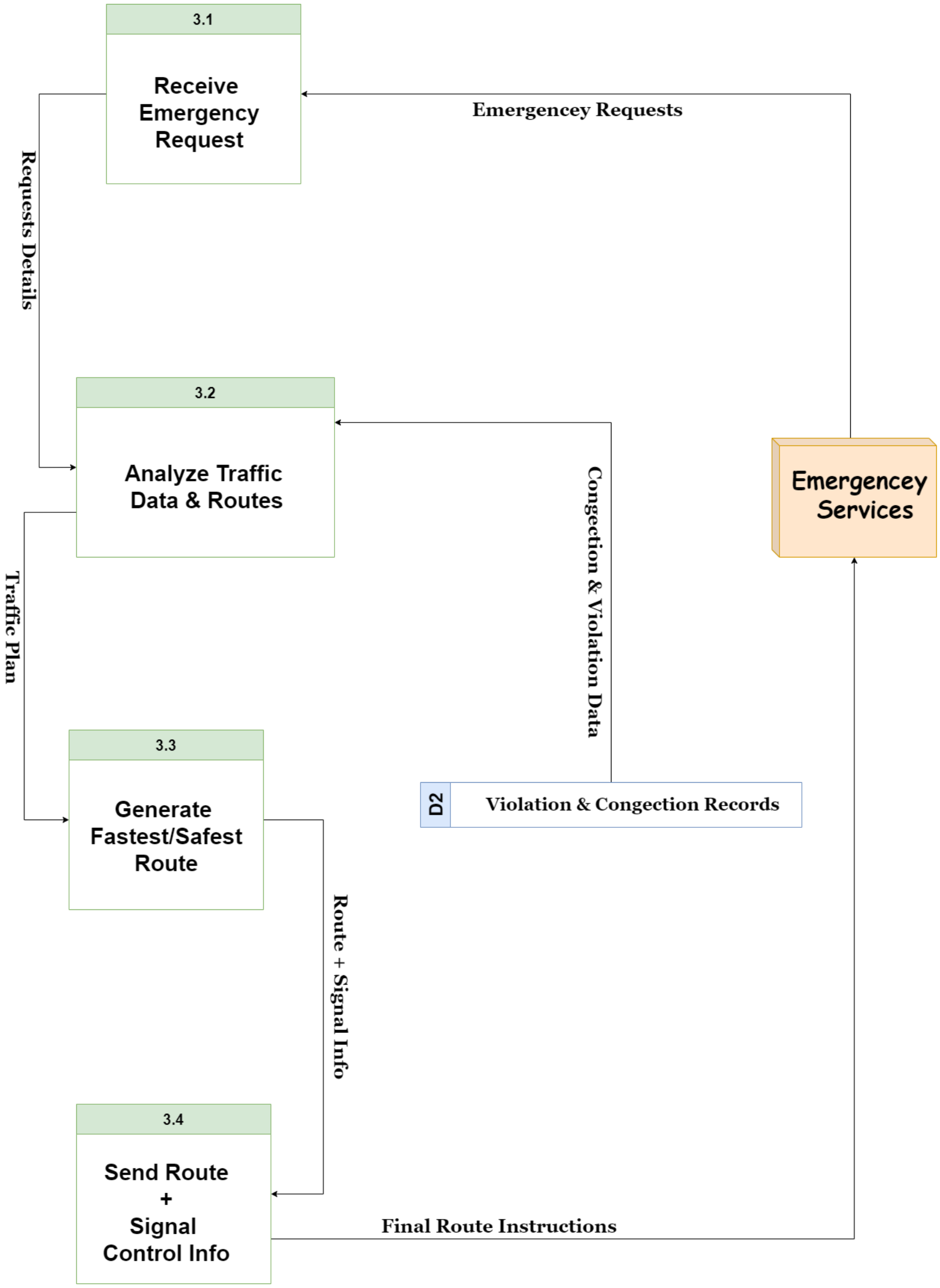
Process 1 :



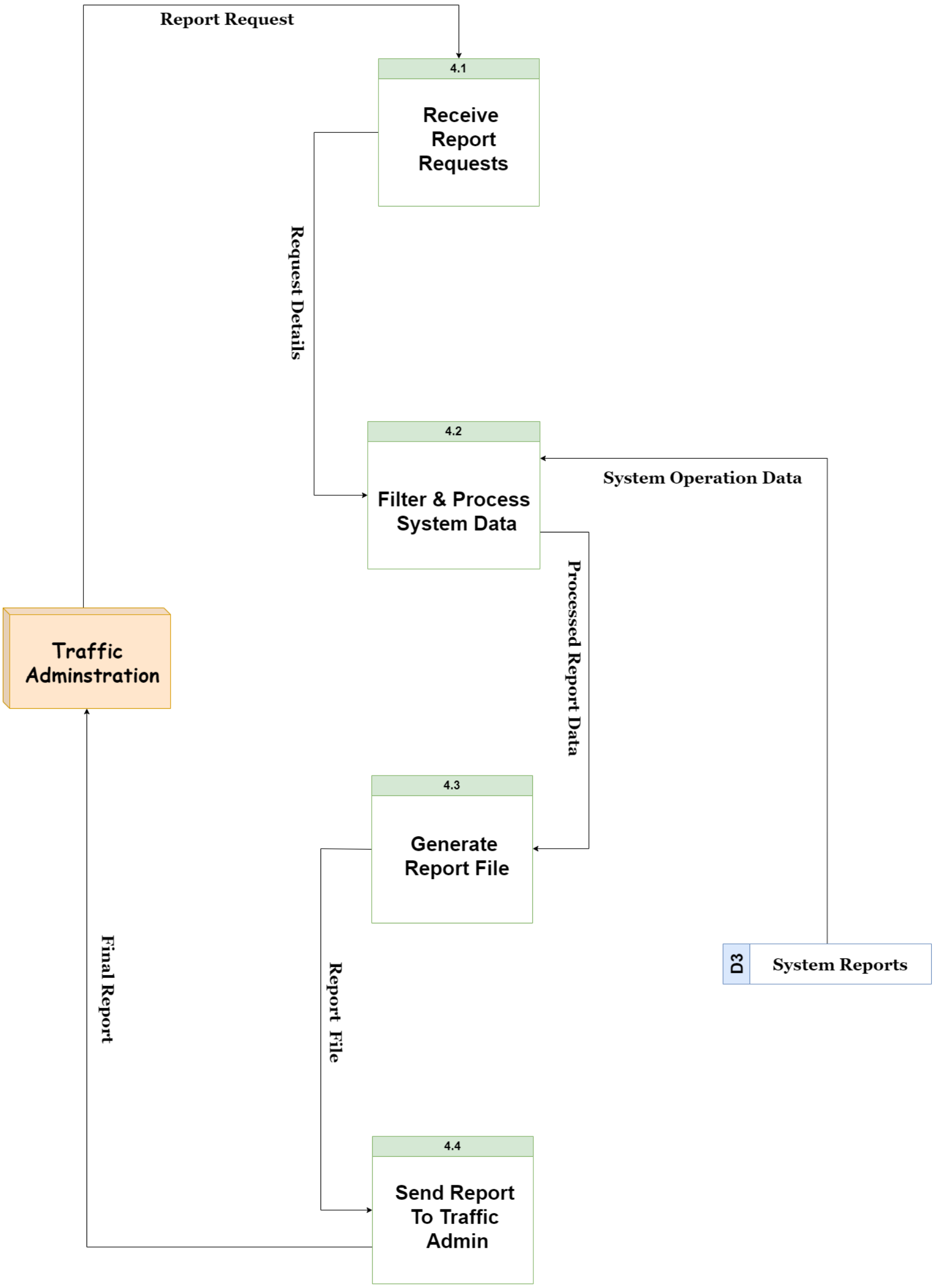
Process 2 :


Violation Congection Info
Traffic Police
2.1
Receive Violation & Congection
Save Info
Raw Violation Data
D2
Violation & Congection Records
2.2
Violation Confirmation
Confirmation Request
Store Confirmed
Confirmed Violation + Comments
Violation Result
Sensor
Check Violation
2.3
Check Driver Status
Violation Status
Status Decision
2.4
Send Alerts & Routes To Driver
Driver ID
Drivers
Alerts & Best Routes
2.5
Send Violation summery
Violation Summary
Read Violation Records
4
Generate Reports

Process 3 :


3.1
Receive Emergency Request
Emergencey Requests
Requests Details
3.2
Analyze Traffic Data & Routes
Emergencey Services
Congection & Violation Data
Traffic Plan
3.3
Generate Fastest/Safest Route
D2
Violation & Congection Records
Route + Signal Info
3.4
Send Route + Signal Control Info
Final Route Instructions

Process 4 :


Report Request
4.1
Receive Report Requests
Request Details
4.2
Filter & Process System Data
System Operation Data
Processed Report Data
Traffic Adminstration
4.3
Generate Report File
Final Report
Report File
D3
System Reports
4.4
Send Report To Traffic Admin

Data Dictionary

1) Context diagram

Process			
Process Name	**Input Data-Flow**	**Output Data-Flow**	**Description**
Smart Traffic Manager	- Location details, Personal data (from Drivers) - Reports Requests, Driver & Vehicle data (from Traffic Administrations) - Check Violation Request (from Sensors) - Comments to violation (from Traffic Police) - Priority Request (from Emergency Services)	- Violation Congestion details (to Drivers) - Performance & Violation reports (to Traffic Administrations) - Violation Result (to Sensors) - Violation & Congestion reports (to Traffic Police) - Fastest roads & paved routes (to Emergency Services)	The core processing unit of the system. It gathers and integrates real-time data from various external entities such as drivers, sensors, traffic police, and administrations. The system analyzes traffic conditions, violations, and emergencies, then distributes relevant reports, requests, and feedback to the appropriate entities to optimize urban mobility and ensure efficient traffic management.

External Entities			
Entities Name	**Input Data-Flow**	**Outpot Data-Flow**	**Description**
Drivers	Violation details & Congestion details	Location details Personal data	Drivers provide violation and congestion data to the system and receive updates on their location and personal information management.
Traffic Adminsrations	Performance and Violation reports	Reports Requests details , Driver and Vechicle data	Receive analytics and raw data for evaluation and policy-making, and send reports for system logging.
Traffic Police	Violation Reports , Congestion reports	Comments to violation	Report traffic issues and violations to the system and receive contextual feedback or comments on cases.
Sensor	Check Violation Request	Violation Result	Detect traffic violations based on requests from the system.
Emergency Services	Fastest Roads	Proprity Request	Receive optimized routing data for emergency responses and send high-priority route requests.

Data Flow			
Data Flow	**Source**	**Destination**	**Description**
Location details	Drivers	Traffic Manager System	Driver's location data sent to the system for monitoring and traffic assessment.
Congestion details	Drivers	Traffic Manager System	Drivers report traffic congestion to the system.
Personal data	Drivers	Traffic Manager System	Driver's personal information like license number sent to the system.
Violation details	Traffic Manager System	Drivers	System sends traffic violation details to the drivers.

Congestion reports	Traffic Manager System	Traffic Police	System sends congestion reports to the traffic police.
Violation Reports	Traffic Manager System	Traffic Police	Traffic violation reports shared with the police for enforcement.
Comments to violation	Traffic Police	Traffic Manager System	Police send feedback or updates regarding violations to the system.
Proprity Request	Emergency Services	Traffic Manager System	Emergency units request priority routes from the system.
Fastest Roads	Traffic Manager System	Emergency Services	System sends the fastest available roads for emergency response.
Check Violation Request	Traffic Manager System	Sensor	System asks sensors to check for any violations.
Violation Result	Sensor	Traffic Manager System	Sensors return results indicating whether a violation occurred.
Performance & Violation reports	Traffic Manager System	Traffic Administrations	Reports on traffic performance and violations sent to administrators.
Reports Requests	Traffic Administrations	Traffic Manager System	Administrators request specific reports from the system.
Vechile & Driver Info	Traffic Administrations	Traffic Manager System	Vehicle and driver data provided to the system by the administration.

2) DFD 0 diagram Data dictionary

Process			
Name	**Inputs**	**Outputs**	**Description**
1. Data Collection	Driver Info, Vehicle Info	Save Driver Data	Collect and store driver and vehicle data from drivers and Traffic Administration.
2. Process Violation & Congestion	Driver & Vehicle Info, Confirmation Request, Check Violation Request	Violation Reports, Congestion Alerts (Best Routes), Save Violations	Analyze driver data, check violations, send alerts, and confirm violations.
3. Handle Emergency	Violation & Congestion Records	Priority Request, Fastest Routes	Process congestion data to help emergency services with route planning.
4. Generate Reports	Violation Summary, Requests	Reports, Save Reports	Generate system and administrative reports based on system activity.

Entities			
Name	**Inputs**	**Outputs**	**Description**
Drivers	Congestion Alerts (Best Routes)	Driver Info	Provide driver data and receive best route suggestions.
Traffic Police	Violation Reports, Confirmation Request	Confirmed Violation + Comments	Review and confirm violations with comments.
Sensor	Check Violation Request	Violation Result	Check if driver has violations and allow/deny passage.
Traffic Administration	Reports	Vehicle Info, Requests	Provide vehicle data and request system reports.
Emergency Services	Priority Request	Fastest Routes	Receive prioritized route planning data in emergencies.

Data Store

Name	Inputs	Outputs	Description
D1. Driver & Vehicle Info	Save Driver & Vehicle Info	Driver & Vehicle Info	Store for validated driver and vehicle data.
D2. Violation & Congestion Records	Save Violations	Read Congestion Info	Stores violation and congestion data for analysis.
D3. System Reports	Save Reports	System Reports	Stores generated reports for system use.

Data Flow

Data Flow	Source	Destination	Description
Save Driver Data	Traffic Police	Data Collection	Data flow from Traffic Police to Data Collection containing: Save Driver Data
Confirmation Request	Traffic Police	Process Violation & Congestion	Data flow from Traffic Police to Process Violation & Congestion containing: Confirmation Request
Violation Reports	Traffic Police	Process Violation & Congestion	Data flow from Traffic Police to Process Violation & Congestion containing: Violation Reports
Driver Info	Drivers	Data Collection	Data flow from Drivers to Data Collection containing: Driver Info
Driver Data	Data Collection	Process Violation & Congestion	Data flow from Data Collection to Process Violation & Congestion containing: Driver Data
Congestion Alerts (Best Routes)	Process Violation & Congestion	Drivers	Data flow from Process Violation & Congestion to Drivers containing: Congestion Alerts (Best Routes)
Check Violation Request	Process Violation & Congestion	Sensor	Data flow from Process Violation & Congestion to Sensor containing: Check Violation Request

Save Violations	Process Violation & Congestion	Violation & Congestion Records	Data flow from Process Violation & Congestion to Violation & Congestion Records containing: Save Violations
Confirmed Violation + Comments	Process Violation & Congestion	Traffic Police	Data flow from Process Violation & Congestion to Traffic Police containing: Confirmed Violation + Comments
Violation Summary	Process Violation & Congestion	Traffic Administration	Data flow from Process Violation & Congestion to Traffic Administration containing: Violation Summary
Violation Result	Sensor	Process Violation & Congestion	Data flow from Sensor to Process Violation & Congestion containing: Violation Result
Read Congestion Info	Violation & Congestion Records	Handle Emergency	Data flow from Violation & Congestion Records to Handle Emergency containing: Read Congestion Info
Priority Request	Handle Emergency	Emergency Services	Data flow from Handle Emergency to Emergency Services containing: Priority Request
Fastest Routes	Emergency Services	Handle Emergency	Data flow from Emergency Services to Handle Emergency containing: Fastest Routes
Requests	Traffic Administration	Generate Reports	Data flow from Traffic Administration to Generate Reports containing: Requests
Reports	Generate Reports	Traffic Administration	Data flow from Generate Reports to Traffic Administration containing: Reports
Save Reports	Generate Reports	System Reports	Data flow from Generate Reports to System Reports containing: Save Reports

Process Description

Violation and Congestion Processing

Description

Handles detection and management of traffic violations and congestion to ensure smooth traffic flow and enforcement of regulations.

Steps

1. Receive driver and vehicle data from the system.
2. Collect violation data from road sensors and surveillance.
3. Send violation confirmation requests to traffic police.
4. Analyze congestion data from sensors.
5. Generate and send congestion alerts and alternate route suggestions to drivers.
6. Store verified violation data into the database.
7. Generate and forward violation reports to traffic authorities.

Inputs

- Driver and vehicle information
- Violation data from sensors
- Congestion data
- Confirmation requests from traffic police

Outputs

- Violation reports to traffic police
- Congestion alerts and route suggestions to drivers
- Stored violation records
- Violation summaries

Emergency Handling

Description

Manages priority routing for emergency vehicles to reduce response time and improve public safety.

Steps

8. Receive emergency request from emergency services.
9. Analyze real-time congestion data to determine fastest route.
10. Check for any violations or road blocks affecting the route.
11. Generate and send optimized route to emergency services.
12. Log emergency response and routing information into system.

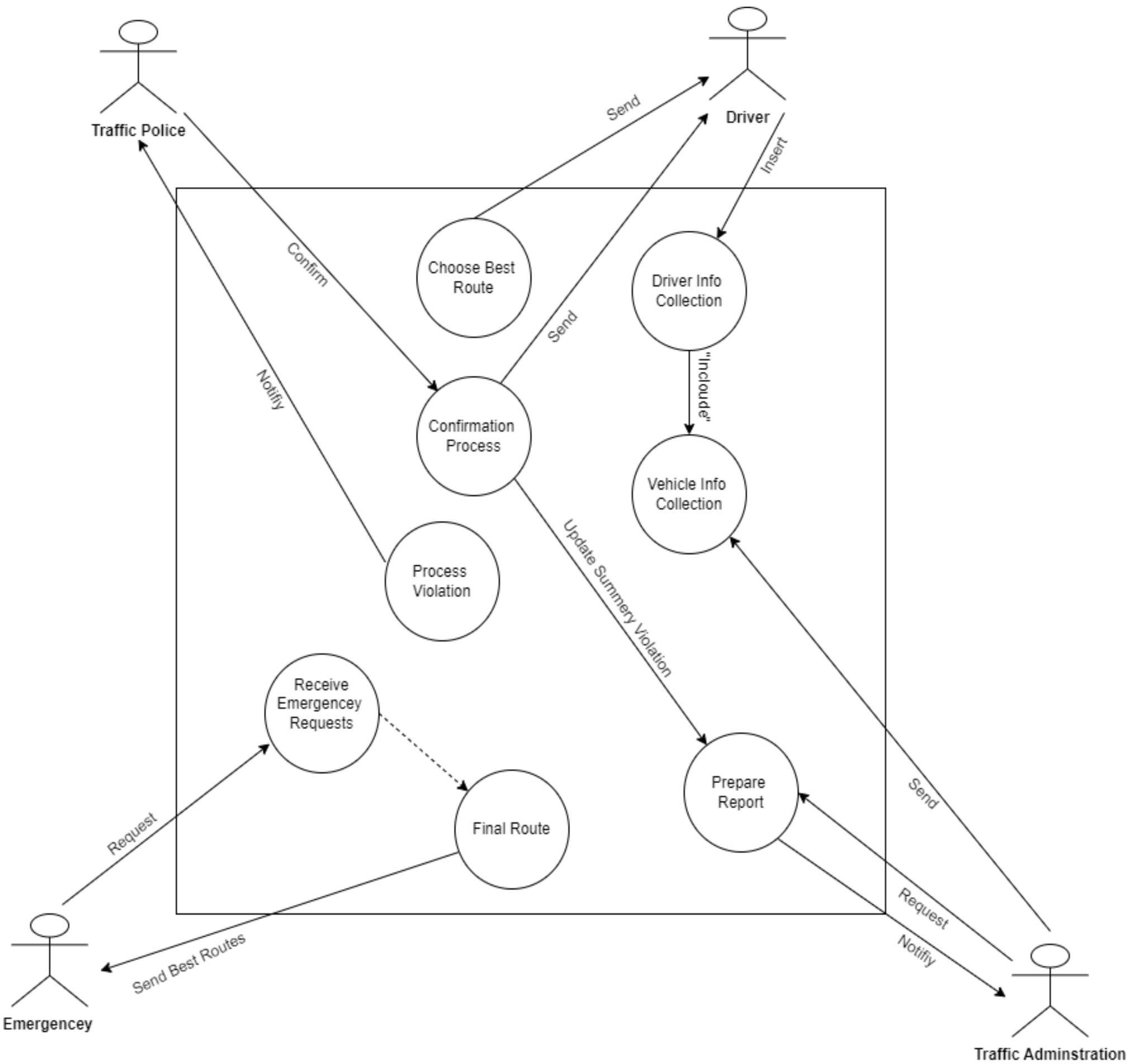
Inputs

- Emergency service requests
- Real-time congestion data

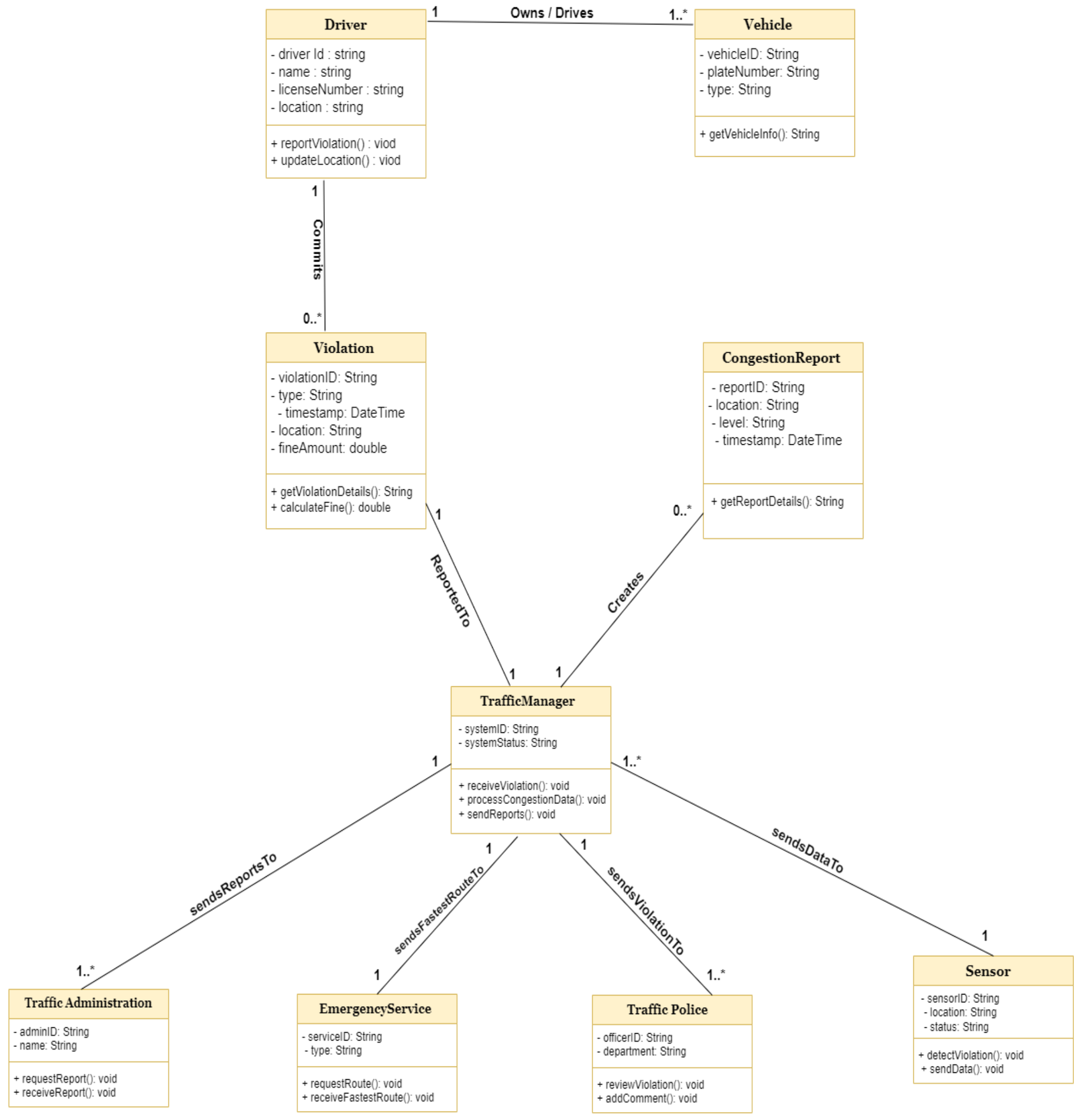
Outputs

- Optimized emergency route suggestions
- Emergency handling logs
- Violation summaries if applicable

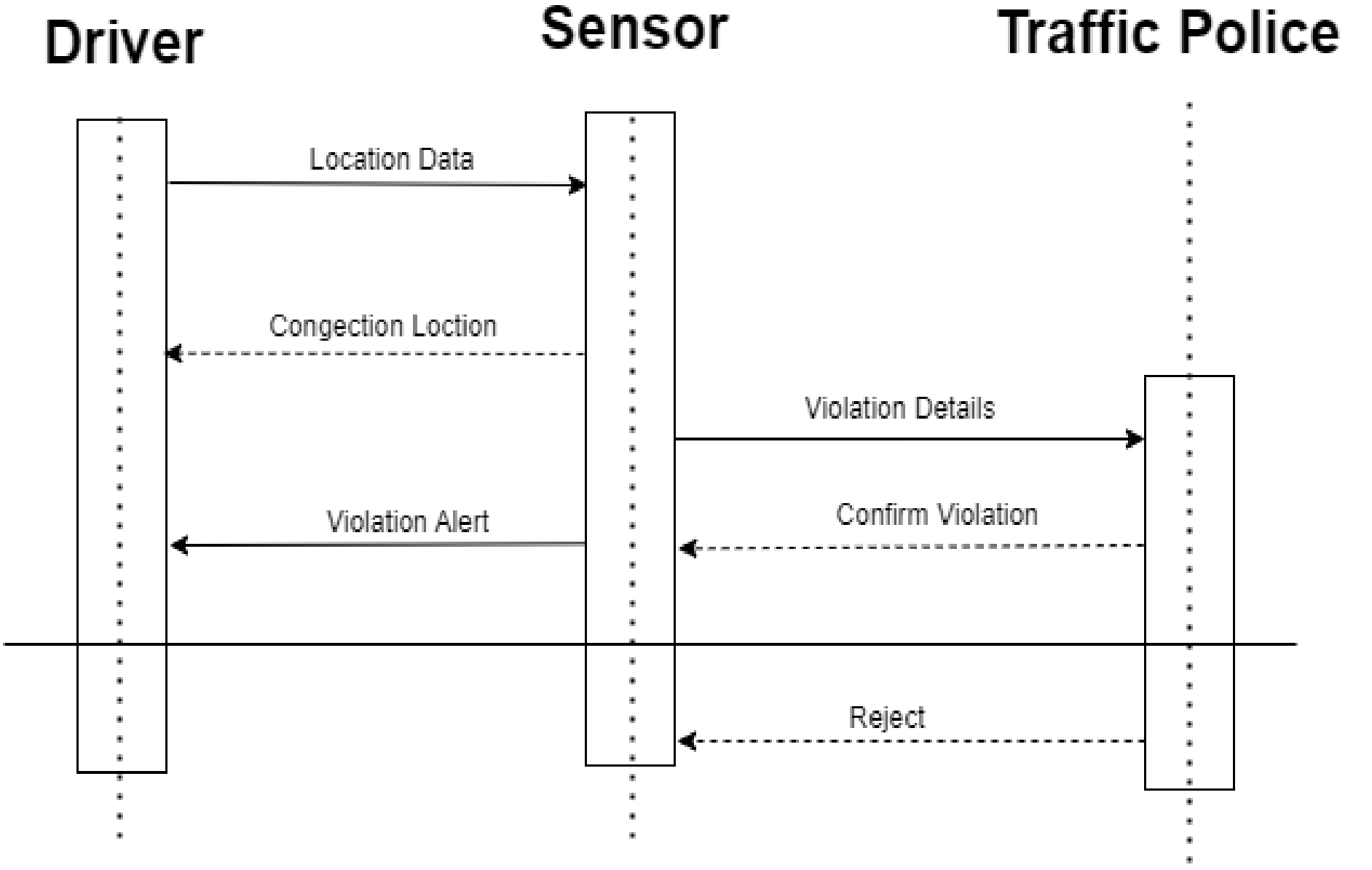
UseCase Diagram



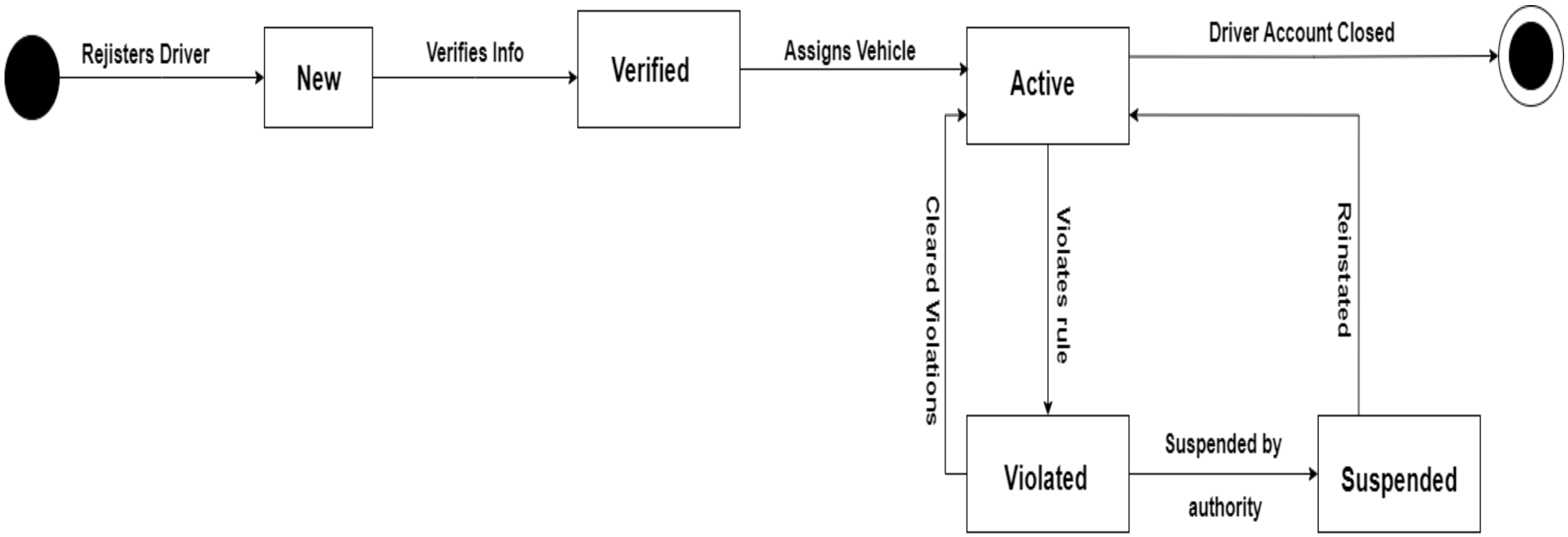
Class Diagram



Sequence Diagram



Transactio Diagram



Activity Diagram

